

# Project Proposal

## Credit Card Fraud Detection

### Using Machine Learning

**Program:** BS Artificial Intelligence, FAST-NUCES Karachi

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#### 1. Introduction

Credit card fraud is a persistent problem in the financial industry, causing billions of dollars in losses annually. Financial institutions rely on machine learning systems to detect fraudulent transactions in real time while minimizing false alarms that inconvenience legitimate customers. This project aims to design, implement, and evaluate a machine learning pipeline for credit card fraud detection, following practices used in the financial industry for handling imbalanced data, model evaluation, and interpretability.

#### 2. Problem Statement

Fraudulent transactions represent an extremely small fraction of all transactions (typically well below 1%), making fraud detection a highly imbalanced classification problem. Standard classification approaches and evaluation metrics such as accuracy are inadequate in this setting, since a model that never predicts fraud can still achieve over 99% accuracy while being practically useless. This project addresses the challenge of building a model that reliably identifies fraudulent transactions while managing the tradeoff between false positives and false negatives.

#### 3. Objectives

- To analyze and understand the characteristics of imbalanced transaction data.
- To apply and compare techniques for handling class imbalance.
- To build and evaluate baseline and advanced classification models for fraud detection.
- To evaluate models using metrics appropriate for imbalanced classification.
- To interpret model predictions using explainability techniques.
- To optionally demonstrate the model through a real-time prediction interface.

## 4. Dataset

- **Source:** Kaggle — Credit Card Fraud Detection (ULB dataset)
- **Link:** <https://www.kaggle.com/mlg-ulb/creditcardfraud>
- **Description:** The dataset contains 284,807 transactions made by European cardholders, of which 492 are fraudulent (0.17%). Due to confidentiality, most features (V1–V28) are the result of a PCA transformation. The dataset also includes Time and Amount as non-transformed features, along with the binary Class label indicating fraud.

## 5. Methodology

### 5.1 Exploratory Data Analysis

The dataset will first be examined to confirm the extent of class imbalance and to understand feature distributions. This includes analyzing the distribution of transaction Amount and Time for fraudulent versus legitimate transactions, and examining feature correlations.

### 5.2 Handling Class Imbalance

To address the severe class imbalance, multiple techniques will be applied and compared, including Synthetic Minority Oversampling Technique (SMOTE), class weighting (e.g., `scale_pos_weight` in XGBoost), and undersampling of the majority class. To prevent data leakage, the train-test split will be performed prior to any resampling.

### 5.3 Model Development

Models will be developed in increasing order of complexity:

1. **Baseline Model:** Logistic Regression, selected for its speed and interpretability.
2. **Primary Model:** XGBoost or LightGBM, with hyperparameters tuned for imbalanced classification.
3. **Extended Analysis (Stretch Goal):** Unsupervised anomaly detection using Isolation Forest or Autoencoders, reflecting techniques used by fraud detection teams to identify previously unseen fraud patterns.

### 5.4 Model Evaluation

Given the severe class imbalance, accuracy will not be used as an evaluation metric. Instead, the models will be evaluated using Precision, Recall, F1-score, and Precision-Recall AUC (PR-AUC), which is more informative than ROC-AUC in imbalanced settings. A

confusion matrix and precision-recall curve will be plotted for the final model. The classification threshold will be selected based on the business tradeoff between missed fraud cases and false alarms, and this tradeoff will be explicitly discussed in the final report.

### **5.5 Model Explainability**

SHAP (SHapley Additive exPlanations) values will be computed for the final model to identify which features contribute most significantly to fraud predictions. This step is included to reflect real-world industry requirements, where model interpretability is essential for regulatory compliance and analyst trust.

### **5.6 Deployment (Optional Extension)**

As an optional extension, the trained model will be deployed as a REST API using FastAPI, capable of returning a fraud probability score for a given transaction. A simple Streamlit dashboard may be built on top of this API to demonstrate real-time predictions.

## **6. Tools and Technologies**

- **Programming Language:** Python
- **Data Processing:** pandas, NumPy
- **Machine Learning:** scikit-learn, XGBoost, LightGBM, imbalanced-learn
- **Explainability:** SHAP
- **Deployment (optional):** FastAPI, Streamlit

## **7. Expected Deliverables**

- A complete exploratory data analysis report.
- Implementation of baseline and primary classification models.
- A comparative evaluation report of imbalance-handling techniques and models.
- A model explainability report using SHAP.
- (Optional) A working demonstration through a deployed API and dashboard.

## **8. Significance**

This project reflects a real-world machine learning pipeline as practiced in the financial technology industry, encompassing data imbalance handling, appropriate evaluation methodology, and model interpretability. It is designed to demonstrate not only technical

implementation skills but also an understanding of the practical considerations required to deploy machine learning responsibly in a high-stakes financial context.